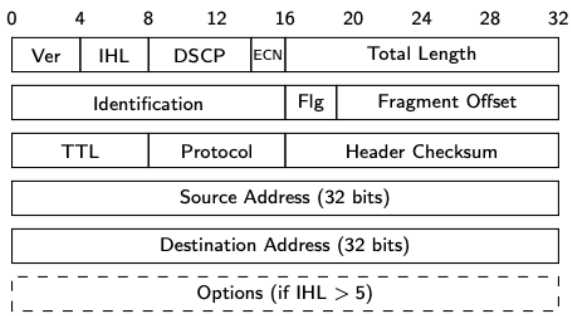
Lecture 17 - IPv4, IPv6, NAT, ICMP

Acronym	Meaning
CIDR	Classless Inter-Domain Routing
DF	Don't Fragment
DHCP	Dynamic Host Configuration Protocol
DSCP	Differentiated Services Code Point
ECN	Explicit Congestion Notification
ICMP	Internet Control Message Protocol
IHL	Internet Header Length

Acronym	Meaning
MTU	Maximum Transmission Unit
NAT	Network Address Translation
NDP	Neighbor Discovery Protocol
PAT	Port Address Translation
PMTUD	Path MTU Discovery
SLAAC	Stateless Address Auto-Configuration
TTL	Time to Live

IPv4 header



what routers examine at each hop:

- Destination address → forwarding decision
- TTL → decrement, drop if 0
- Header checksum → verify and recompute
- Total length → packet boundary

What passes through

- Source address
- Protocol → demux key (1=ICMP, 6=TCP, 17=UDP)
- DSCP
- Payload

Key Fields: TTL and Protocol

<u>TTL</u> → time to live (8 bits)

- Each router decrements by 1
- If TTL → 0 ; drop packet, send <u>ICMP Time Exceeded</u>
- Prevents packets from looping forever in routing loops

<u>Protocol</u> → 8 bits

- Tells the destination host which upper layer module should receive the payload → the <u>demltiplexing key</u>
- Analogous to the EtherType field in ethernet frames

DSCP: IP's QoS Mechanism

The internet is fundamentally best effort. DSCP provides a way to differentiate traffic without per flow signaling

DSCP	Per-Hop Behavior	Typical Use	Priority
000000	Best Effort (default)	Web browsing	Lowest
001010–011010	Assured Forwarding	Video streaming	Medium
101110	Expedited Forwarding	VoIP calls	Highest

How DSCP works

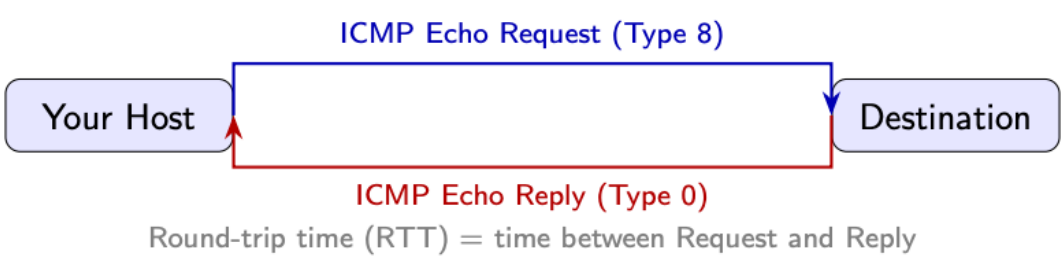
The sender (or edge router) **marks** packets with a DSCP value. Each router independently decides how to treat each class — prioritize in queues, drop lower-priority traffic first during congestion. **No per-flow state.** Scales to the core internet.

ICMP: The Internet's Error Reporter

- Internet Control Message Protocol (ICMP) provides error reporting and diagnostics for IP
- It is part of the network layer but carried inside IP datagrams (protocol = 1)
- For error messages: body includes the original IP header + first 8 bytes of the offending packet

Type	Code	Name	Purpose
0	0	Echo Reply	Response to ping
3	0	Dest Unreachable: Net	No route to network
3	1	Dest Unreachable: Host	Can reach net, not host
3	3	Dest Unreachable: Port	Host reached, port closed
3	4	Frag Needed, DF Set	Packet too big, can't fragment
8	0	Echo Request	Ping request
11	0	Time Exceeded: TTL	TTL decremented to 0

How ping Works - At the Protocol Level



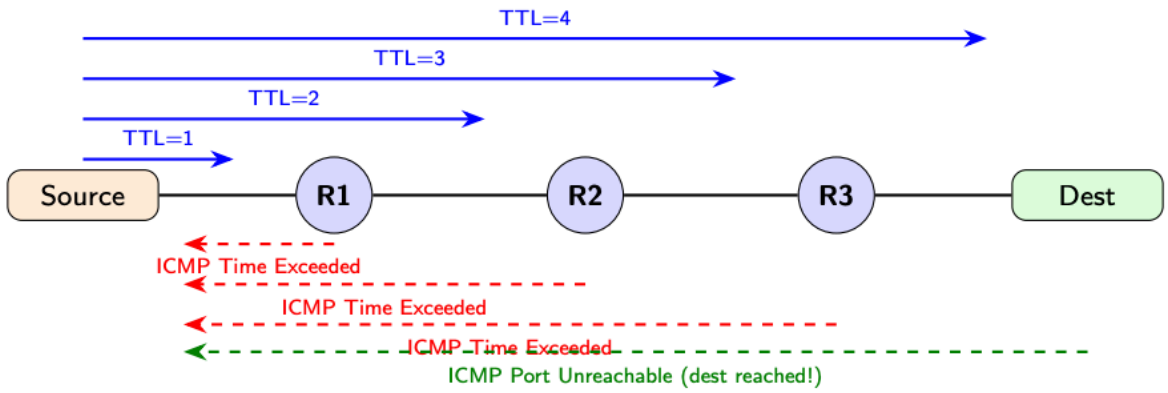
1. Host cerates ICMP Echo Request with identifier + sequence number
2. Encapsulated in IP datagram with protocol = 1

3. Routed to destination through the forwarding machinery
4. Destination replies with ICMP Echo Reply
5. Sender Measures RTT

Key insight

`ping` does **not** use TCP or UDP. It operates directly at the network layer using ICMP.

IP Fragementation



1. Send apckets with TTL = 1 → first router sends back ICMP Time Exceeded
2. Increment TTL each round → discover each router along the path
3. Destination responds with ICMP Port Uncreachable (Unix) or Echo Reply (Widnows)

In output = that router does not

Why Fragmentation is Problematic

Problem:

1. Losing one fragment = Losing te entire diagram
   - other fragments wasted bandwidth
2. Reassembly is complex
   - buggering , ordering, timeout handling at destination
3. Security vulnerabilities
   - Overlapping fragemrnts exploited in attacks
4. Header overhead
   - Each fragment carries its own 20 byte IP header

Solution:

1. Sender sets DF (dont fragment) flag
2. If packets too big, router sends, ICMP fragmentation Needed including the link MTU
3. Sender reduces packet size and retries
4. Repeat until the path MTU is found

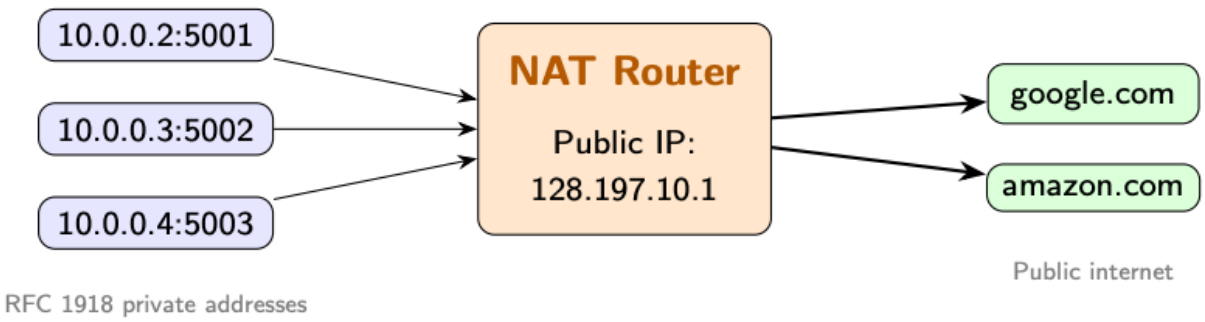
Caution

Firewalls that block all ICMP break PMTUD — creates "black holes" where large packets silently disappear.

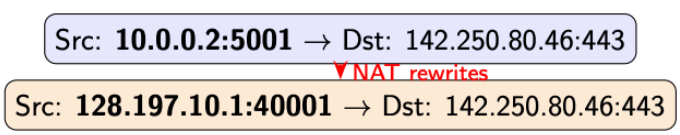
NAT: The Hack that Saved the Internet

Problem: IPv4 has 2^{23} = 4.3 billion addresses but far more devices

Solution: network address translation (NAT) → an entire private network shares one public IP address → NAT replaces the source IP and port on outgoing packets, records the mapping, and restores them on replies



NAT Translation Table



NAT Table

WAN Side	LAN Side
128.197.10.1:**40001**	10.0.0.2:5001
128.197.10.1:**40002**	10.0.0.3:5002
128.197.10.1:**40003**	10.0.0.4:5003

Port Address Translation (PAT) → the most common form

- A single public IP supports ip to 65,000 simultaneous connectios
- NAT modifies both the IP header addresses and transport header (ports)
- Must resimpute IP and TCP/UDP checksums

NAT and the END-END Principle → NAT fundamentally violates the end to end priciple

What breaks:

- Incoming conenctions fails
  - External hosts cannot reach private addresses
- Peer to peer is complicated
  - Two hosts behind NAT cannot directly communicate
- Embedded IPs breaks
  - Protocols like FTP, SIP, embed addresses in payload

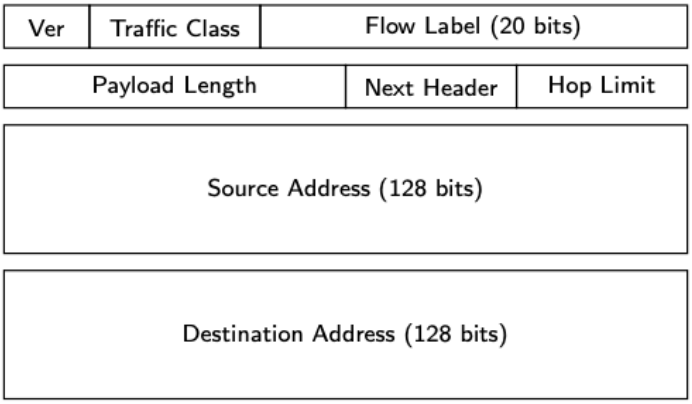
Work arounds:
- Port forwarding (manual)
- UPnP / NAT-PMP (automatic)
- STUN/TURN/ICE (WebRTC, VoIP)
- Hole punching (gaming, P2P)

IPv6 : the original version (1990s) → Designed to solve the IPv4 address exhaustion crisis
- Vast address space: 128 bits → 2 128 ≈ 3.4 × 1038 addresses
- Global addressability: every device gets a real, routable address → no NAT
- Simplified header: fixed 40 bytes, no checksum, faster router processing
- No fragmentation in transit: only endpoints fragment
- Auto-configuration (SLAAC): hosts configure themselves without DHCP
- Mandatory IPsec: built-in encryption for every connection

IPv6: The reality check
Why adoption stalled:
- NAT extended IPv4's life by 20 years
- Enterprise: IPv4 + NAT works fine
    - Upgrade cost > perceived benefit
- Dual stack infrastructure is expensnive
- Applications need testing on IPv6
- If its not broken, dont fix

Who actually deployed IPv6
- Mobile carriers
    - Tmobile, verizon, etc
- Phone likely has a real global IPv6 address not NAT

IPv6: Header: Simplified



Fixed 40 bytes → no variable length options in the base header
What changes from IPv4:
- Removed: header checksum (redundant), fragmentation fields, IHL options
- Renamed: TTL → Hop limit, protocol → next header

- Added: flow label (20 bits) for identifying packet flows
- Optional features moved to extension headers

IPv4 vs IPv6: side by side

Feature	**IPv4**	**IPv6**
Address size	32 bits (4.3B addresses)	128 bits (3.4×10^{38})
Header size	20–60 bytes (variable)	40 bytes (fixed)
Header checksum	Yes (recomputed every hop)	**Removed**
Fragmentation	In base header; routers fragment	**Removed**; only source fragments
Options	In base header (variable IHL)	Extension headers
TTL / Hop Limit	Time to Live	Hop Limit (renamed)
Broadcast	Yes (255.255.255.255)	**No broadcast** — multicast only
Auto-config	DHCP required	SLAAC (stateless)

Why remove the header checksum?

Recomputing it at every hop wastes CPU. Link-layer CRCs catch bit errors per link. Transport-layer checksums (TCP/UDP) catch end-to-end corruption. The IP header checksum was **redundant work** at every router.

IPv6 Address notation

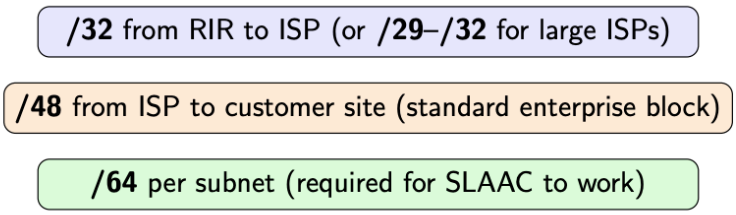
Full form: 8 groups of 4 hex digits, separated by colons: `2001:0db8:85a3:0000:0000:8a2e:0370:7334`

Compression rules → applied in order:

1. Drop leading zeros in each group
    a. 0db8 → dbb
    b. 0370 → 370
    c. 0000 → 0
    d. 2001:db8:85a3:0:0:8a2e:370:7334
2. Replace one longest run of consecutive all 0 groups with ::
    a. 2001:db8:85a3::8a2e:370:7334

IPv6 address types and allocation

Type	Prefix	Scope	Purpose
Global Unicast	`2000::/3`	Internet	Globally routable (= public IPv4)
Link-Local	`fe80::/10`	Single link	Auto-configured; neighbor discovery
Unique Local	`fc00::/7`	Organization	Like RFC 1918 private addresses
Multicast	`ff00::/8`	Varies	**Replaces broadcast**
Loopback	`::1/128`	Host	Equivalent to 127.0.0.1



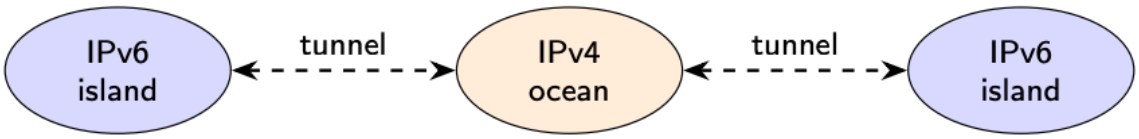
Allocation hiearcrchy:

IPv6 Transition Mechanisms

Dual stack → hosts run both IPv4 and IPv6. DNS returns both A and AAAA records
→ happy eyeballs algorithms prefers IPv6 when available

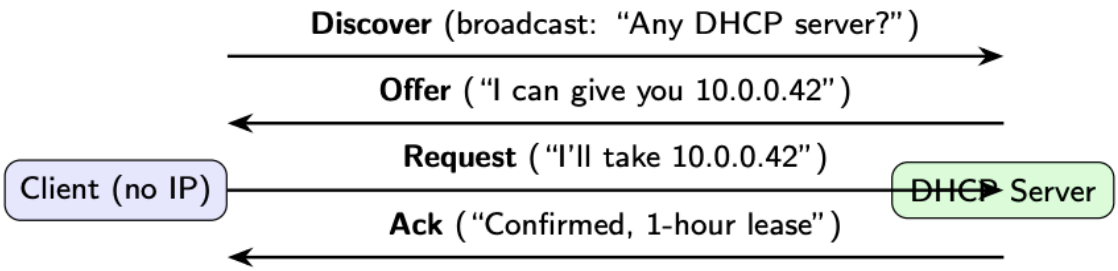
Tunneling → IPv6 packets encapsulated inside IPv4 only segements
→ IPv6 islands communicate across IPv4 ocean

NAT64/ DNS64 → translates between IPv6 only clients and IPv4 only servers
→ maps IPv4 addresses into a special IPv6 prefix



DHCP: how hosts get their addresses

problem : a new host has no IP, no subnet mask, no gateway no DNS server



- Uses UDP (port 67 server, port 68 client) → can't use TCP with no IP address
- Client sends from 0.0.0.0 to 255.255.255.255 (broadcast)
- Leases expire → client must renew or address returns to pool
- Relay agents: routers forward DHCP broadcasts across subnets

SUMARY:

Topic	Key Takeaway
IPv4 header	20-byte base with TTL, Protocol, addresses; DSCP for QoS
ICMP	Error reporting (Dest Unreachable, Time Exceeded) and diagnostics (ping, traceroute) — the protocol behind L13's demos
Fragmentation	Problematic — PMTUD avoids it; IPv6 bans router fragmentation
NAT	Saved IPv4 but broke end-to-end; uses port translation
IPv6	128-bit addresses, simplified header, no broadcast; mobile-driven adoption
DHCP	Discover/Offer/Request/Ack — how hosts bootstrap onto a network